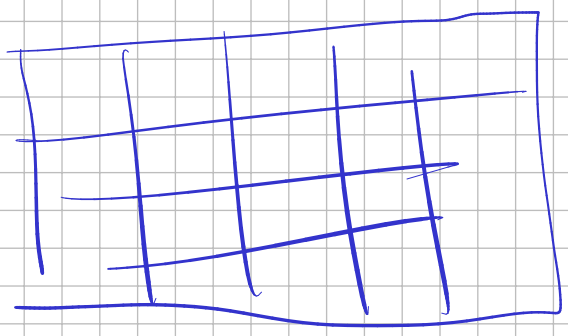
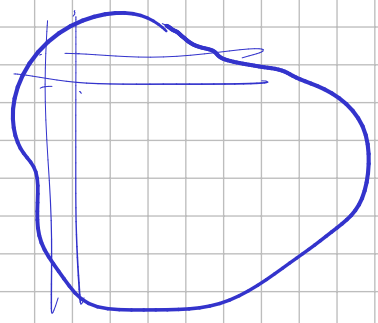




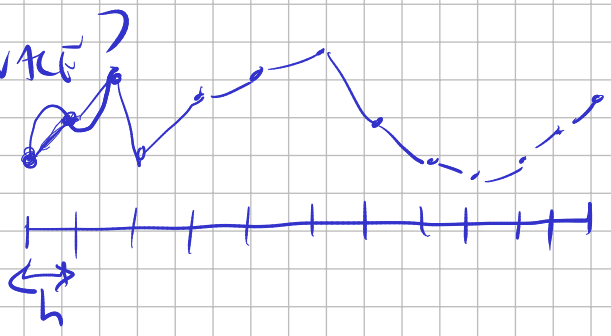
PROBLEMI ELLITTICI IN FORMA DEBOLE

- $a(u,v) = F(v)$

TROVA $u \in V$: $\forall v \in V$ VALE

$V = H^1$

- POSSIAMO TROVARE UN SURROGATO DI V che sia FINITO DIMENSIONALE?



- $V_h \subset V \subset H^1(\Omega)$

$V_h \xrightarrow[h \to 0]{} V$

$a_h : V_h \times V_h \to \mathbb{R}$ BILINEARE CONTINUA COERCIVA

$F_h : V_h \to \mathbb{R}$ LINEARE, CONTINUA

$a_h(u_h, v_h) := a(u_h, v_h) \quad \forall u_h, v_h \in V_h \subset V$

$F_h(u_h) := F(u_h) \quad \forall u_h \in V_h$

TROVARE $u_h \in V_h$: $\forall v_h \in V_h$

$a(u_h, v_h) = F(v_h)$

PROBLEMA DI GALERKIN

$N_h = \dim(V_h) < +\infty$ $\qquad \mathbb{R}^n = \left\langle \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \dots \right\rangle$

SCEGLIAMO N_h FUNZIONI DI BASE DI V_h $\quad \{\varphi_i\}_{i=1}^{N_h}$ $\quad \varphi_i(x)$ $\quad \varphi_i \in V$, $\in V_h$

$\forall u_h \in V_h \Rightarrow u_h(x) = \sum_{j=1}^{N_h} u_j \varphi_j(x)$

$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$ $\quad$ (TROVA u_h)

$\Updownarrow$

$a(u_h, \underline{\varphi_i}) = F(\underline{\varphi_i}) \quad \forall \varphi_i \quad i = 1, \dots, N_h$

$a\left(\sum_{j=1}^{N_h} u_j \cdot \varphi_j, \varphi_i\right) = F(\varphi_i) \quad \forall i = 1, \dots, N_h$

(TROVA u_j $j = 1, \dots, N_h$)

$\sum_{j=1}^{N_h} \underbrace{a(\varphi_j, \varphi_i)} \cdot u_j = \underbrace{F(\varphi_i)} \quad \forall i = 1, \dots, N_h$

$a_{ij} := a(\varphi_j, \varphi_i) \qquad f_i := F(\varphi_i)$

$$\boxed{\underline{\underline{A}}\, \underline{u} = f}$$

$\underline{\underline{A}} \in \mathbb{R}^{N_h \times N_h} \qquad f, \underline{u} \in \mathbb{R}^{N_h}$

a : COERCIVO, CONTINUO, BILINEARE

A MATRICE $\Rightarrow A$ è DEFINITA POSITIVA

- A è DEF. POS se $\boxed{v^T A v \geq 0}$ $\forall v \in \mathbb{R}^{N_h}$

e $v^T A v = 0 \iff v = 0 \in \mathbb{R}^{N_h}$

$\underline{v} = \{v_i\}_{i=1}^{N_h} \iff v_h = \sum_{j=1}^{N_h} v_j \varphi_j \in V_h$

$$v^T A v = \sum_{j=1}^{N_h} \sum_{i=1}^{N_h} v_i a_{ij} v_j = \sum_{i,j} v_i a(\varphi_j, \varphi_i) \cdot v_j$$

$$= \sum_j a\left(\varphi_j, \sum_i v_i \varphi_i\right) v_j = a\left(\sum_j v_j \varphi_j, \sum_i v_i \varphi_i\right)$$

$$= a(v_h, v_h) \geq \alpha \cdot \|v_h\|_V^2 \geq 0 \qquad \alpha > 0$$

$$v^T A v = 0 \iff \alpha \|v_h\|_V^2 = 0 \iff \|v_h\|_V = 0$$

$$v_h \equiv 0 \implies v_i = 0 \quad \forall i = 1, \dots, N_h. \quad \boxtimes$$

- $\exists!$ SOL PROBLEMA DI GALERKIN
- STABILE ?
- $u_h \xrightarrow[h \to 0]{} u$

- TEOREMA LAX-MILGRAM SPAZIO DI HILBERT V_h

$\implies a(\cdot,\cdot)$ CONT, COER, SYM, BIL

$F(\cdot)$ LINEARE, CONT

$\implies \boxed{\exists!}$ u_h $a(u_h, v_h) = F(v_h)$ $\forall v_h \in V_h$.

- <u>STABILITÀ</u> $\|u_h\|_V \leq \frac{1}{\alpha} \|F\|_{V^*}$

α è COSTANTE DI COERCIVITÀ DI a

$$a(u,u) \geq \alpha \|u\|_V$$

$$\underbrace{\|u_h\|_V^2}_{} = \|u_h\|_{V_h}^2 \leq \frac{1}{\alpha} a(u_h, u_h) = \frac{1}{\alpha} F(u_h)$$

$$\underbrace{\leq \frac{1}{\alpha} \|F\|_{V^*} \cdot \|u_h\|_V}_{}$$

$$\|u_h\|_V \leq \frac{1}{\alpha} \|F\|_{V^*} .$$

$$u_h : \quad a(u_h, v_h) = F(v_h) \qquad \forall v_h \in V_h$$

$$w_h : \quad a(u_h, v_h) = G(v_h) \qquad \forall v_h \in V_h$$

$$\underbrace{\|u_h - w_h\|_V}_{} \leq \frac{1}{\alpha} \underbrace{\|F - G\|_{V^*}}_{}$$

<u>CONVERGENZA</u> SUPPONIAMO CHE $h \to 0$

$u_h \to u \in V$

$$\lim_{h \to 0} \underset{v_h \in \underline{\underline{V_h}}}{\arg\min} \|v_h - u\|_V = 0$$

u è UN PROBLEMA SCALARE

1) $a(u_h, v_h) = F(v_h) \qquad \forall v_h \in V_h \subset V$

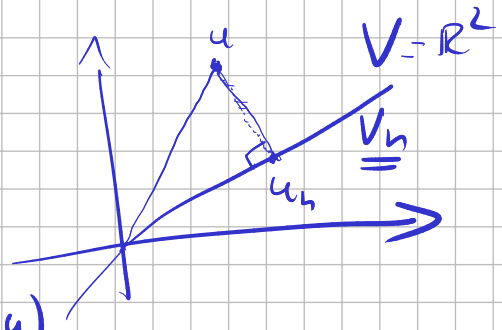
2) $a(u, v) = F(v) \qquad \forall v \in V$

3) $a(u, v_h) = F(v_h) \qquad \forall v_h \in V_h \in V$

1) - 3) $\Rightarrow \quad a(u_h, v_h) - a(u, v_h) = \cancel{F(v_h)} - \cancel{F(v_h)} = 0$

$$\to \boxed{a(u_h - u, v_h) = 0} \qquad \forall v_h \in \underline{\underline{V_h}}$$

LEMMA DI CÉA



$\forall v_h \in V_h$

$$a(u-u_h, u-u_h) =$$

$$= \underbrace{a(u-u_h, u-v_h) + a(u-u_h, \underbrace{v_h - u_h}_{\in V_h})}_{=0}$$

PER ORTOGONALITÀ DI GALERKIN

$$|a(u-u_h, u-u_h)| = |a(u-u_h, u-v_h)| \le C \cdot \|u-u_h\|_V \cdot \|u-v_h\|_V$$

$\forall v_h \in V_h$

$$\| \quad a(u-u_h, u-u_h) \ge \alpha \|u-u_h\|_V^2$$

$$\|u-u_h\|_V \le \frac{C}{\alpha} \cdot \|u-v_h\|_V \qquad \forall v_h \in V_h$$

$$\|u-u_h\|_V \le \frac{C}{\alpha} \cdot \underbrace{\inf_{v_h \in V_h} \|u-v_h\|_V}_{\text{MIGLIORE APPROX IN } V_h} \to 0$$

$$\downarrow$$

$$0$$

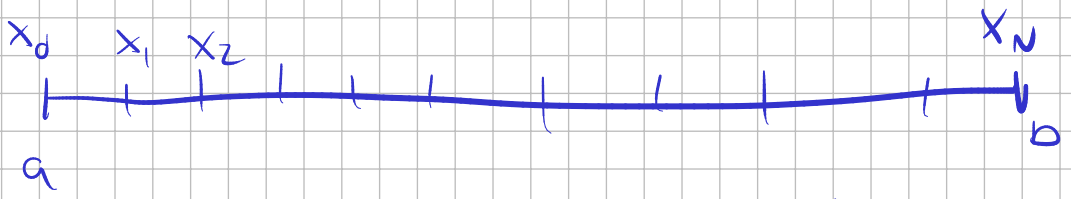
Se $\lim_{h\to 0} \inf_{v_h \in V_h} \|u-v_h\| = 0 \Rightarrow \lim_{h\to 0} \|u-u_h\|_V = 0$

C e α NON DIPENDONO DA V_h MA SOLO DA a!

Se $\inf_{v_h \in V_h} \|u-v_h\|_V = O(h^p) \Rightarrow \|u-u_h\|_V = O(h^p)$

$\boxtimes$

ELEMENTI FINITI IN 1D



$$a = x_0 < x_1 < \dots < x_N = b \qquad \mathcal{T} := \bigcup_{j=1}^{N} K_j$$

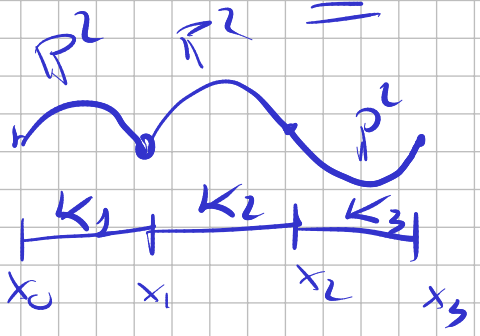
$$K_j := (x_{j-1}, x_j) \qquad j = 1, \dots, N$$

$$h = \max_{j=1,\dots,N} h_j \qquad h_j = x_j - x_{j-1}$$

ELEMENTI FINITI CONTINUI

$$X_h^r := \{ v_h \in \underline{\underline{C^0(\bar{\Omega})}} : v_h|_{K_j} \in \mathbb{P}^r(K_j) \ \forall K_j \in \mathcal{T} \}$$

$$\subset H^1((a,b))$$



$X_h^1 \leftarrow$ LINEARI $\mathbb{P}^1$

$\forall\, j = 1, \dots, N$ elemento $\mathbb{P}^1$ 2 POSSIBILITÀ

$2 \cdot N$ GRADI DI LIBERTÀ

$N - 1$ VINCOLI DI CONTINUITÀ

$\Rightarrow 2N - (N-1) = N+1$ GRADI DI LIBERTÀ

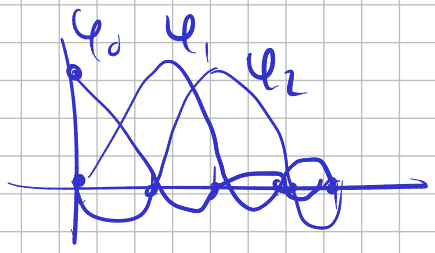
$$\dim(X_h^1) = \underline{\underline{N+1}}$$

QUALE BASE? LAGRANGIANO $\Rightarrow$ COEFFICIENTI

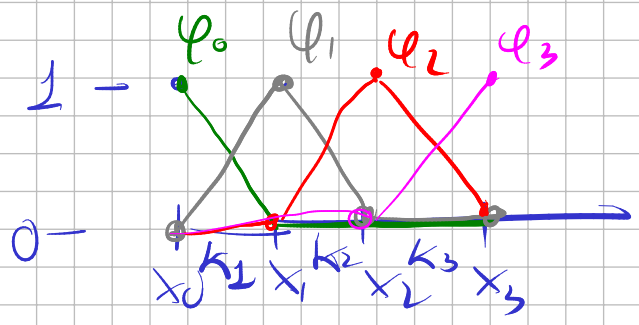
SONO I VALORI AD ALCUNI PUNTI

$\rightsquigarrow \quad V(x_i) = \underbrace{V_i}_{\text{DOF}}$ BASI?

$$V(x) = \sum_{i=1}^{N+1} V_i \, \varphi_i(x)$$



POLINOMI NODALI LAGRANGIANI



PROPRIETÀ

$$\varphi_i(x_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$\Rightarrow \varphi_i(x) = \begin{cases} \dfrac{x - x_{i-1}}{x_i - x_{i-1}} & x_{i-1} < x \leq x_i \\[2ex] \dfrac{x_{i+1} - x}{x_{i+1} - x_i} & x_i < x < x_{i+1} \\[2ex] 0 & \text{else} \end{cases}$$

$$\mathrm{supp}(\varphi_i) = [x_{i-1}, x_{i+1}]$$

$$K_i \cup K_{i+1}$$

$$a_{ij} = a(\varphi_j, \varphi_i) \neq 0 \iff j \in \{i-1, i, i+1\}$$

$\downarrow$

MATRICE SPARSA ☺

es. $a(u,v) = \int_a^b u' \cdot v' \, dx$

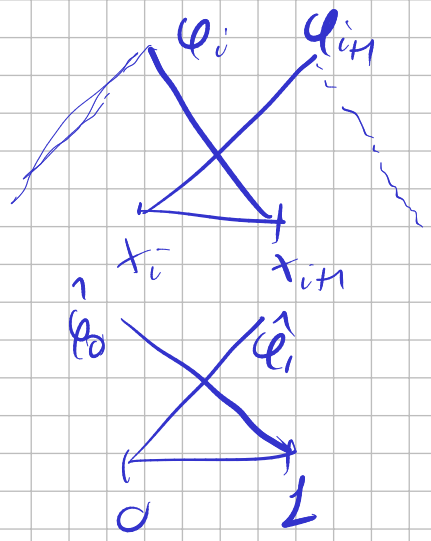
ELEMENTO DI RIFERIMENTO $[0,1]$ =

es. ELEMENTO FISICO $K_{i+1} = [x_i, x_{i+1}]$

T_{i+1} TRASFORMAZIONE AFFINE (LINEARE)

$$T_{i+1} : [0,1] \to K_{i+1}$$

$$x = T_{i+1}(\xi) = x_i + \xi \cdot \underbrace{(x_{i+1} - x_i)}_{h_{i+1}}$$



$$\xi = T_{i+1}^{-1}(x) = \frac{x - x_i}{x_{i+1} - x_i}$$

$$\hat{\varphi}_0(\xi) := \varphi_i(T_{i+1}(\xi))$$
$$= 1 - \xi$$

$$\hat{\varphi}_1(\xi) = \xi$$

$$\varphi_i(x) = \hat{\varphi}_0\left(T_{i+1}^{-1}(x)\right) = \hat{\varphi}_0\left(\frac{x - x_i}{x_{i+1} - x_i}\right)$$

$\boxed{\mathbb{P}^1}$

$\mathbb{P}^2$? $\quad X_h^2$

$$\hat{\varphi}_0(\xi) = 2\left(\xi - \tfrac{1}{2}\right)(\xi - 1)$$

$$\hat{\varphi}_1(\xi) = 4\xi(1 - \xi)$$

$$\hat{\varphi}_2(\xi) = 2\xi\left(\xi - \tfrac{1}{2}\right)$$

es. $\mathbb{P}^2$

$$\varphi_2(x) = \begin{cases} \hat{\varphi}_2(T_1^{-1}(x)) & x \in K_1 \\ \hat{\varphi}_0(T_2^{-1}(x)) & x \in K_2 \\ 0 & \text{else} \end{cases}$$

$$\varphi_3(x) = \begin{cases} \hat{\varphi}_1(T_2^{-1}(x)) & x \in K_2 \\ 0 & \text{else} \end{cases}$$

$$\varphi_{2i}(x) = \begin{cases} \hat{\varphi}_2(T_i^{-1}(x)) & x \in K_i \\ \hat{\varphi}_0(T_{i+1}^{-1}(x)) & x \in K_{i+1} \end{cases}$$

$$\varphi_{2i+1} = \hat{\varphi}_1(T_i^{-1}(x)) \quad x \in K_i$$